

***A list comprehension is a compact way of defining a list in one line:***

```
>>> nums = [10**exp for exp in range(5)]  
>>> nums  
[1, 10, 100, 1000, 10000]
```

A list comprehension uses square brackets, an expression that generates each item in the list, and a `for` loop as the basis for the list.

A list comprehension can modify each item in an existing list:

```
>>> states = ['CA', 'CO', 'CT']  
>>> lower_states = [state.lower()  
...     for state in states]  
>>> lower_states  
['ca', 'co', 'ct']
```